

EE1101: Circuits and Network Analysis

Assignment - 10

Handed out: 18 - Oct - 2024

Due : 28 - Oct - 2024 (before 5 PM)

Instructions :

1. Please upload your assignment solutions to the course page on the Canvas platform. Only solutions submitted through canvas will be reviewed. For specific guidelines, refer to the instructions provided on the course page.
2. It is suggested that you attempt all the problems. However, it is sufficient to submit solutions for problems that total 10 points.
3. Submissions received after the deadline will attract negative marking. Ensure that your submissions are named in the following format: RollNo-Assignment-10.pdf.

1. (6 points) Consider the circuit shown in Fig. 1¹.

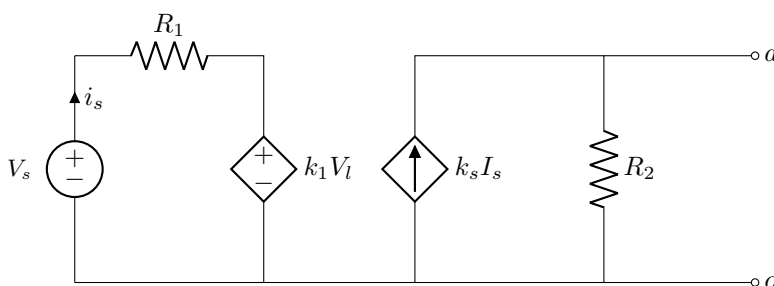


Fig. 1: Circuit for Question 1

- (a) (2 points) Compute the Thevenin equivalent circuit at the port $a-o$.
 - (b) (2 points) Compute the Norton equivalent circuit at the port $a-o$.
 - (c) (2 points) If a variable resistance (say R_l) is connected at the port $a-o$, compute the value of R_l at which maximum power is delivered to R_l .
2. (6 points) Consider the circuit shown in Fig. 2.
 - (a) (2 points) Compute the Thevenin equivalent circuit at the port $a-o$.
 - (b) (2 points) Compute the Norton equivalent circuit at the port $a-o$.
 - (c) (2 points) If a variable resistance (say R_l) is connected at the port $a-o$, compute the value of R_l at which maximum power is delivered to R_l .
 3. (8 points) For the circuit shown in Fig. 3, compute the Thevenin and Norton equivalent circuit at the port $a-o$.

¹Questions 1-2 are from the reference: M. E. Van Valkenburg, "Network Analysis"

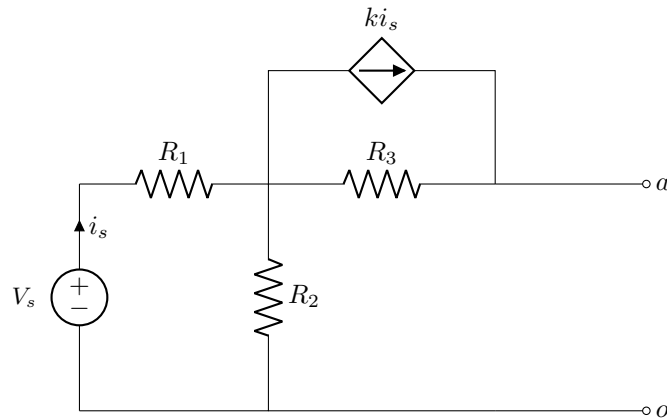


Fig. 2: Circuit for Question 3

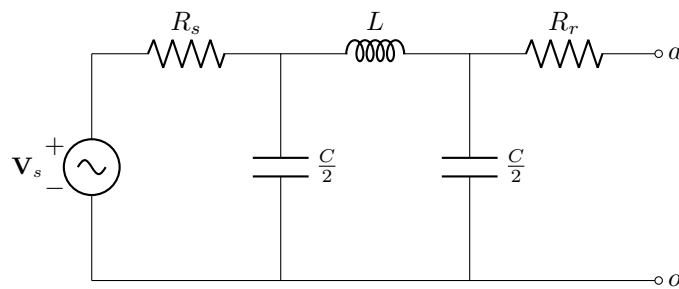


Fig. 3: Circuit for Question 3

- (a) (2 points) Compute the Thevenin equivalent circuit at the port $a-o$.
 - (b) (2 points) Compute the Norton equivalent circuit at the port $a-o$.
 - (c) (2 points) If a variable resistance (say R_l) is connected at the port $a-o$, compute the value of R_l at which maximum power is delivered to R_l .
 - (d) (2 points) If a variable impedance (say $\mathbf{z}_l$) is connected at the port $a-o$, compute the value of $\mathbf{z}_l$ at which maximum power is delivered.
4. (2 points) For the circuit shown in Fig. 4, compute the admittance matrix of the circuit. Consider the nodes as indicated in the figure.

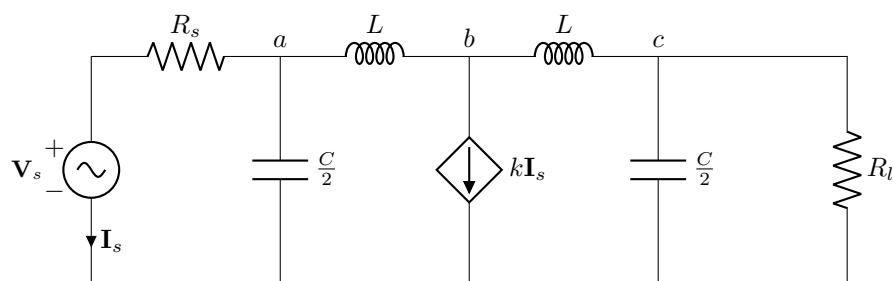


Fig. 4: Circuit for Question 4

5. (2 points) Represent the node equations of the circuit shown in Fig. 5 in the form $\mathbf{Y}\mathbf{V} = \mathbf{V}$ where $\mathbf{V} = [\mathbf{V}_a, \mathbf{V}_b, \mathbf{V}_c]^t$.

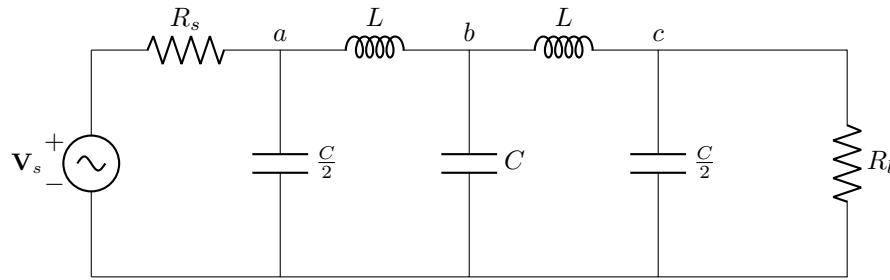


Fig. 5: Circuit for Question 5

6. (2 points) For the circuit shown in Fig. 6, derive an expression for the steady state voltage across the capacitor using superposition theorem.

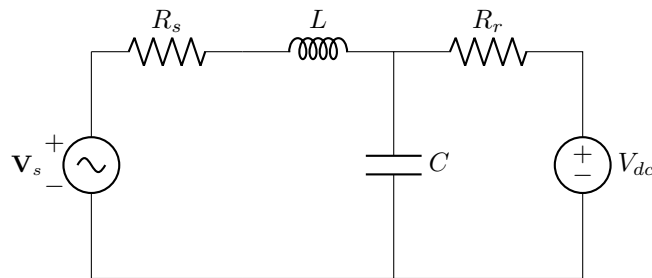


Fig. 6: Circuit for Question 6

7. (2 points) For the circuit shown in Fig. 7, derive an expression for the steady state voltage across the capacitor when $v(t) = V_m [\cos \omega t + \frac{1}{3} \cos 3\omega t]$.

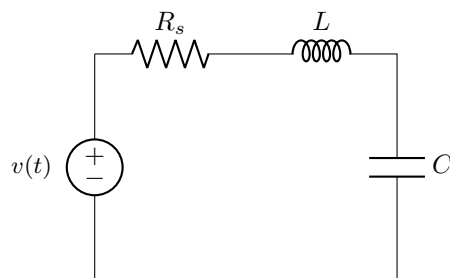


Fig. 7: Circuit for Question 7